

# Design Document – Zombie Dog Platformer

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## Project Summary

This project is a competitive wave-based platformer where the player controls a dog fighting zombies, collecting items, and surviving increasingly difficult levels. The game matters because it provides fast-paced challenge, replayability, and entertainment for younger audiences who enjoy skill-based games that steadily increase in difficulty.

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## Problem Statement

Many teenagers get bored of simple or repetitive games, so this project aims to provide a challenging, engaging platformer experience that keeps players motivated through harder levels, enemies, and mechanics.

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## Use Case

Teenagers and young gamers will use this product because it offers fast-paced competitive gameplay, difficult levels, and rewarding progression. They will play by navigating platforms, fighting zombies, collecting power-ups, and completing all 25 levels.

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## Goals and Objectives

1. Create a visually appealing, challenging platformer with smooth player movement and intuitive controls.
  2. Develop increasing difficulty through harder enemies, time limits, platform layouts, and item decision-making.
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## Key Features and Functions

1. **Wave-based platformer levels** that become significantly harder after Level 5.
  2. **Combat mechanics** including melee sword attacks, guns with limited ammo, and enemy types with unique behaviors.
  3. **Item system** with potions, dashes, shields, double jumps, and two-slot inventory for strategic gameplay.
  4. **Multiple enemy classes** (slow heavy, fast weak, big tank) that require different strategies.
  5. **Keys, quests, and collectibles** to progress through each level.
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## Tech Stack and Tools

- **Game Engine:** Unity or Godot
  - **Programming Language:** C# (Unity) or GDScript (Godot)
  - **Art/Animation:** Aseprite, Photoshop, or Piskel
  - **Sound/Music:** Bfxr, Audacity
  - **Version Control:** Git + GitHub
  - **Project Management:** Trello or Notion
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## Algorithm (High-Level Functionality)

1. Start game → Display menu
2. Player selects “Start” → Load Level 1
3. Spawn player, enemies, items, and platforms
4. While level is active:
  - a. Player moves and fights enemies
  - b. Player collects coins, items, and keys
  - c. Check health, time limit, and enemy waves
5. If key is found and all enemies defeated → unlock exit
6. Player enters exit → Increase difficulty → Load next level

7. Repeat until Level 25 or player dies

8. Show results screen

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## Flowchart (High-Level)

Start



Main Menu



Load Level



Spawn Player + Enemies



[Decision] Player Survives?

- **No** → Game Over

- **Yes** → Continue



Find Key + Defeat Enemies



Level Exit Unlocked



Advance to Harder Level



Repeat until Level 25



End / Victory Screen

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# **Timeline (Milestones)**

## **Month 1 – Core Development**

- Build movement system
- Fix sliding/“slideshow” issue
- Add jump, double jump, and dash logic

## **Month 2 – Combat + Enemies**

- Implement sword, gun, and item mechanics
- Add slow, fast, and big zombie types
- Add health bar + item bar

## **Month 3 – Level Design**

- Create 25 levels
- Add keys, quests, timers, collectibles

## **Month 4 – Polish + Playtesting**

- Improve visuals & animations
- Smooth movement

- Implement feedback (difficulty, spikes, more enemies)
  - Final bug fixes
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## Risk Mitigation

**Risk:** The game may become too easy or repetitive, leading to player boredom.

**Mitigation:** Add scaling difficulty, new enemy attacks, tighter time limits, harder platforming, and randomized enemy patterns on higher levels.

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## Evaluation Criteria (Measures of Success)

1. Players report increased challenge and improved gameplay difficulty after the first 5 levels.
  2. Smooth, intuitive controls that feel responsive and polished.
  3. Playtesters rate the final game at **4/5 or higher**, matching early prototype results.
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## Future Considerations

**Maintenance Need:** Continuous bug fixes for item interactions, enemy AI, and physics.

**Future Feature:** Add an online leaderboard or endless mode so players can compete for high scores.

